

MODEL 705-1A MICRO FACE (cont.)

The "Execute" command, which is an output signal from the Micro Face, indicates that the Micro Face is ready to receive numeric data. The "Execute" command is a d.c. condition of 0 or +5 volts with a convenient slide switch to reverse logic levels. The "Execute" d.c. condition is available at pin 31 of the input connector (Figure 3). Output impedance is typically 470 ohms and provides for a 10-milliamp sink. (See Figure 2.)

SIGN OF A NUMBER

A one-bit input at pin 17 of the input connector (Figure 3) is used to designate the sign of the number. Signal levels are the same as those of the data input signals. A convenient slide switch is available to reverse the logic levels of the sign bit.

FIXED DECIMAL POINT

Any one of the seven (7) input digits can be wired to generate a decimal point BCD code 12 allowing for a six (6) digit number plus fixed decimal point.

EXAMPLE: If input numbers are 123.456, the fourth digit can be wired for a decimal point (BCD code 12).

FLOATING DECIMAL POINT

The last digit can be wired to indicate the number of places the decimal point should be located within a six (6) digit number.

EXAMPLE If the input number is 123456 x 10⁻³ the range digit can be wired as the 7th digit and the number would read in as 1234563 (where the 3 indicates under program control the scale factor). This feature can be used for automatic ranging.

ADDRESSING THE 705-1 INPUT MICRO FACE

To switch control from a programmable calculator, a two-step command must be issued either manually from the keyboard or as a program step in core memory.

Addressing the Micro Face requires a Group 1, 07XX (where "XX" can be any number from 00

through 15), thus allowing the operator to choose an address by setting four (4) switches. This feature also allows the use of more than one interface for multiplexing purposes by calling out different addresses. (Because of the IC fan-out restrictions, no more than 10 interfaces are recommended.)

There are eight slide switches conveniently located on the rear cover. Four of these switches (marked "Device Address") generate the "XX" codes 00 through 15 which will be recognized by the selected Micro Face when addressed. The remaining four switches provide the following flexibility as marked.

MARKINGS

Data	Reverse logic levels for input signal
Print	Reverse logic levels for "Print" command
Execute	Reverse logic levels for "Execute" command
Sign	Reverse logic levels for Sign of number

PARALLEL INPUT

The input connector is conveniently wired for direct parallel input of seven digits (28 lines as shown in Figure 3).

Typical entry sequence would be:

Group 1 07XX	2-step command from the programmable calculator transfers control to the selected input interface.
"EXECUTE" ON	D.C. condition changes at pin 31 and remains in that state waiting for a "Print" command (input strobe) from the outside device.
"PRINT"	Command generated from outside device to pin 18 for approximately 5 microseconds or greater. All seven digits are then transferred into the display register of the programmable calculator in approximately 5 milliseconds.
"EXECUTE" OFF	D.C. condition returns to original state and indicates transfer is complete and control is returned to the programmable calculator.